

Week 4 Report

Motivation to Move from ANN/MLP to Graph Neural Networks (GNN)

Introduction

Artificial Neural Networks (ANNs) and Multi-Layer Perceptrons (MLPs) are effective for learning from structured data such as images and tabular datasets. However, they assume that each data sample is independent and cannot model relationships between connected entities. Many real-world datasets, including citation networks, social networks, and molecular graphs, naturally have a graph structure where connections between nodes are important.

Motivation for GNNs

The major limitation of ANN/MLP models is that they ignore graph connectivity and process each node independently. Consequently, valuable information from neighbouring nodes is not utilized, leading to weaker representations. In the Cora citation dataset, for example, research papers are connected through citation links, and these relationships provide important contextual information for classification.

Graph Neural Networks (GNNs) overcome this limitation by learning from both node features and graph structure. Through message passing, each node aggregates information from its neighbours, enabling the model to capture local relationships and improve feature representations.

GCN for Node Classification

In this assignment, a **two-layer Graph Convolutional Network (GCN)** was implemented for node classification on the *Cora citation dataset*. The first GCN layer aggregates information from neighbouring nodes and applies a ReLU activation, while the second GCN layer predicts the class label of each node. Unlike MLPs, GCNs effectively combine node features with graph connectivity, resulting in improved classification performance.

Conclusion

Graph Neural Networks extend traditional neural networks by incorporating relational information between connected nodes. Their ability to exploit both feature and structural information makes them well suited for graph-based applications such as citation analysis, recommendation systems, social networks, and molecular modelling. Consequently, GNNs provide a more effective framework than ANN/MLP for learning from graph-structured data.